COUNTABLE GENERATION DESCENDS ALONG A FAITHFULLY FLAT RING MAP

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Writing elements of a base change as finite sums of pure tensors. Used at the start of the counted proof (lines 22609-22612). Produces the countable set of coefficients whose generated submodule is the candidate for M.
- Faithful flatness as detection of surjectivity after base change. Used at the end of the counted proof (lines 22614-22616). Turns surjectivity of the induced map on base changes into equality M' = M.

Lemma 0.1. *Let $R \to S$ be a faithfully flat ring map. Let M be an R-module. If the S-module $M \otimes_R S$ is countably generated, then M is countably generated.*

Proof. Say $M \otimes_R S$ is generated by the elements y_i, $i = 1, 2, 3, \ldots$. Write $y_i = \sum_{j=1,\ldots,n_i} x_{ij} \otimes s_{ij}$ for some $n_i \geq 0$, $x_{ij} \in M$ and $s_{ij} \in S$. Denote $M' \subset M$ the submodule generated by the countable collection of elements x_{ij}. Then $M' \otimes_R S \to M \otimes_R S$ is surjective as the image contains the generators y_i. Since S is faithfully flat over R we conclude that $M' = M$ as desired. □

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